

EXCEL DATA ANALYTICS PROJECT REPORT

Airline Performance, Data Cleaning & Revenue Analysis

Course: Excel for Data Analysis

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PART A: DATASET INTRODUCTION

- **1. Project Title:** Airline Performance & Revenue Analysis
- **2. Dataset Name:** airline project.xlsx
- **3. Source of Dataset:** Airline Operations & Ticketing Database
- **4. Purpose of Dataset:** To analyze flight schedules, ticketing data, passenger demographics, and revenue to optimize flight operations.
- **5. Number of Records:** 300 raw records (292 clean records)
- **6. Number of Variables:** 7 initial raw variables, expanded to 12 clean variables

7. Important Variables Description:

- **Booking_ID:** Unique identifier for each transaction (e.g., ABK00001)
- **Flight_Date:** Date when the flight is scheduled to operate
- **Route:** Origin and destination city code (e.g., MGQ-NBO, MGQ-ADD)
- **Passenger_Type:** Category of passenger (Adult, Child, Senior)
- **Travel_Class:** Seating class (Economy, Business, Unknown)
- **Ticket_Price:** Price paid for the flight ticket in USD (\$)
- **Booking_Status:** Current status of booking (Completed, Confirmed, Cancelled, Unknown)

PART B: DATA PREPARATION & CLEANING

1. Data Import & Data Types

The raw data was imported from Excel. Each column was checked and formatted with appropriate data types:

- **Text:** Booking_ID, Route, Passenger_Type, Travel_Class, Booking_Status
- **Number / Currency:** Ticket_Price formatted as Currency (\$#,##0.00)
- **Date:** Flight_Date formatted standardly as YYYY-MM-DD

2. Duplicate Records Handling

Duplicate entries reduce metrics accuracy in PivotTables and aggregations. Duplicates were identified using Excel's Highlight Duplicate Rules and removed, resulting in a clean dataset of 292 unique booking records.

3. Missing Values Treatment

Missing values (blank cells and 'Unknown' entries) were handled systematically:

- **Blank Travel_Class / Booking_Status:** Filled with standard categorical placeholder 'Unknown' to maintain record integrity without dropping rows.
- **Missing Ticket_Prices:** Evaluated using average pricing metrics or flagged for manual audit.

PART C: TEXT CLEANING

Demonstrated text standardization using core Excel functions across raw columns:

- **1. TRIM:** TRIM(A2) — Cleared leading/trailing whitespace spaces in text fields.
- **2. PROPER:** PROPER(D2) — Formatted passenger category names into proper casing.
- **3. UPPER:** UPPER(C2) — Ensured route codes were capitalized uniformly (e.g., MGQ-NBO).

PART D: DATA ANALYSIS USING FORMULAS

1. Basic Statistical Formulas

- **=SUM(Ticket_Price):** Calculates total revenue generated across clean bookings (\$58,112.25).
- **=AVERAGE(Ticket_Price):** Determines average price per ticket (\$199.01).
- **=COUNT(Booking_ID):** Counts total confirmed clean bookings (292 records).
- **=MIN(Ticket_Price):** Identifies minimum recorded price.
- **=MAX(Ticket_Price):** Identifies maximum recorded ticket price (\$1,018.00).

2. Conditional Functions

- **=SUMIF(Booking_Status, "Confirmed", Ticket_Price):** Calculates total revenue from confirmed bookings (\$17,272.97).
- **=COUNTIF(Booking_Status, "Completed"):** Counts number of completed flights.
- **=AVERAGEIF(Travel_Class, "Business", Ticket_Price):** Calculates average fare for Business Class (\$318.42).

3. Logical Functions

- **IF Function:** =IF(Ticket_Price >= 300, "Premium", "Standard") — Categorizes bookings into fare brackets.
- **IFS Function:** =IFS(Travel_Class="Business", "High Yield", Travel_Class="Economy", "Standard Yield", TRUE, "Other")

PART E: DATE ANALYSIS, SORTING & FILTERING

1. Date Analysis

- **=YEAR(), =MONTH(), =DAY():** Extracted year, month, and day components for period-over-period reporting.
- **=NETWORKDAYS():** Calculated business duration between flight booking and departure.

2. Sorting & Filtering

- **Sorting:** Sorted routes alphabetically (A-Z) and ticket prices from Largest to Smallest to highlight top revenue transactions.
- **Filtering:** Filtered data to display high-value routes (MGQ-NBO, MGQ-ADD) and evaluated completion status rates.

PART F: CONDITIONAL FORMATTING & PIVOTTABLES

1. Conditional Formatting

- **Color Scales:** Applied color gradients to visually segment low vs. high ticket fares.
- **Highlight Rules:** Highlighted tickets exceeding \$300 in soft green and cancelled bookings in light red.

2. PivotTables Overview

Constructed PivotTables to summarize 292 clean records into actionable metrics across key areas:

- **PivotTable 1 (Revenue by Route):** Rows: Route | Values: Sum of Ticket_Price
- **PivotTable 2 (Class Yield Analysis):** Rows: Travel_Class | Values: Average Ticket_Price
- **PivotTable 3 (Booking Status Summary):** Rows: Booking_Status | Values: Count of Booking_ID, Sum of Ticket_Price

PART G & K: DASHBOARD ARCHITECTURE & KPIS

1. Key Performance Indicators (4 KPI Cards)

- **Total Revenue:** \$58,112.25
- **Total Bookings:** 292 Bookings
- **Average Fare:** \$199.01
- **Completion Rate:** 54.45% (159 completed bookings)

2. Visualizations (Charts & Slicers)

- **Chart 1:** Route Revenue Performance (Bar Chart)
- **Chart 2:** Booking Status Breakdown (Pie/Donut Chart)
- **Chart 3:** Travel Class Yield Comparison (Column Chart)
- **Chart 4:** Monthly Flight Trends (Line Chart)
- **Interactive Slicers:** Interactive Slicers for Route, Travel Class, and Booking Status.

PART M: FINDINGS AND RECOMMENDATIONS

5 Key Findings

- **1. Top Revenue Driver (MGQ-NBO):** Generated \$22,331.80, making it the highest revenue route.
- **2. High Average Fare Route (MGQ-ADD):** Generated \$15,504.33 with the highest average ticket price (\$267.32).
- **3. Premium Class Yield:** Business Class yield averaged \$318.42 compared to Economy Class at \$178.17.
- **4. Completion Rate:** 159 out of 292 bookings (54.45%) completed successfully, generating \$30,899.44.
- **5. Data & Cancellation Gaps:** 30 bookings cancelled (\$5,899.04) and 17 records lacked destination codes.

3 Strategic Recommendations

- **1. Fleet Allocation & Frequency:** Increase seat capacity and flight frequencies on MGQ-NBO and MGQ-ADD routes during peak travel periods.
- **2. Premium Marketing:** Introduce corporate loyalty packages and upgrade promotions to boost high-margin Business Class bookings.
- **3. Operations & Data Integrity:** Enforce validation controls at booking counters to eliminate missing fields and implement automated SMS/email departure reminders to reduce cancellations.